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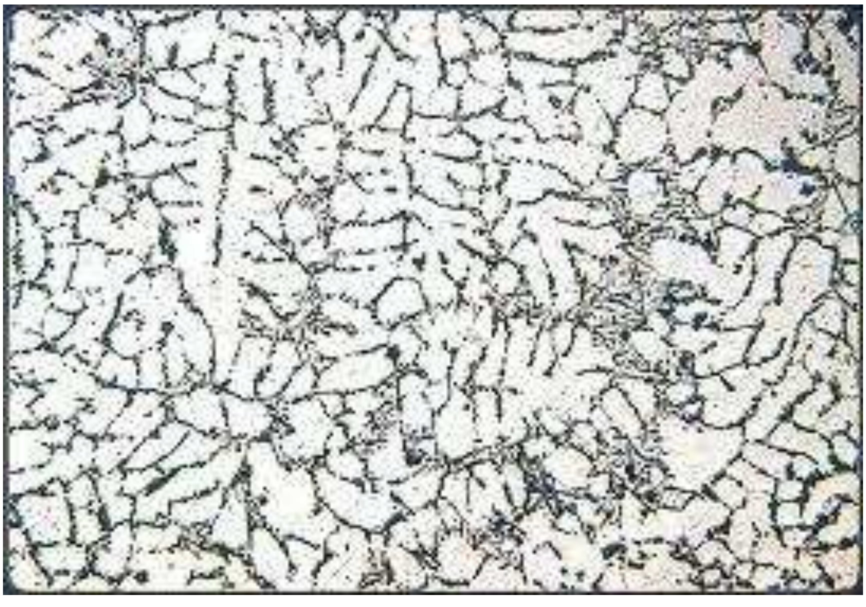
Faculty of Production Technology and Management

FACULTY OF PRODUCTION ENGINEERING AND MANAGEMENT

DEPARMENT OF TECHNOLOGY AND MATERIAL ENGINEERING

PRACTICAL METALLOGRAPHY



30.6.2014

Pasa Yaman

Theoretical Part

What is the basics of preparation of metallographic sample?

Metallography is the study of a material's microstructure. Analysis of a material's microstructure aids in determining if the material has been processed correctly and is therefore a critical step for determining product reliability and for determining why a material failed. The basic steps for proper metallographic sample preparation include: documentation, sectioning and cutting, mounting, planar grinding, rough polishing, final polishing, etching, microscopic analysis and hardness testing.

Documentation – Metallographic analysis is a valuable tool. By properly documenting the initial sample condition and the proceeding microstructural analysis, metallography provides a powerful quality control as well as an invaluable investigative tool.

Sectioning and Cutting – Following proper documentation, most metallographic samples need to be sectioned to the area of interest and for easier handling. Depending upon the material, the sectioning operation can be obtained by abrasive cutting (metals and metal matrix composites), diamond wafer cutting (ceramics, electronics, biomaterials, minerals), or thin sectioning with a microtome (plastics).

Proper sectioning is required to minimize damage, which may alter the microstructure and produce false metallographic characterization. Proper cutting requires the correct selection of abrasive type, bonding, and size; as well as proper cutting speed, load and coolant. Table I list the most common type of abrasive blades used for metallographic sectioning and Table II lists cutting parameters for diamond wafer cutting.

Table I. Abrasive Blade Selection Guidelines Chart

Materials (alloys)	Classification	Abrasive/Bond
Aluminum, brass, zinc, etc.	Soft non-ferrous	SiC/ Rolled rubber
Heat treated alloys	Hard non-ferrous	Alumina/ Rubber resin
<Rc 45 steel	Soft ferrous	Alumina/ Rubber resin
>Rc 45 steel	Hard ferrous	Alumina/ Rubber resin
Super alloys	High Ni-Cr alloys	SiC/ Rolled rubber

Table II. Diamond Wafer Blade Selection Guidelines

Material	Characteristic	Speed (rpm)	Load (grams)	Blade (grit/conc.)
Silicon substrate	soft/brittle	<300	<100	Fine/low
Gallium arsenide	soft/brittle	<200	<100	Fine/low
Boron composites	very brittle	500	250	Fine/low
Ceramic fiber composites	very brittle	1000	500	Fine/low
Glasses	brittle	1000	500	Fine/low
Minerals	friable/brittle	>1500	>500	Fine/low
Alumina ceramic	hard/tough	>1500	>500	Medium / low
Zirconia (PSZ)	hard/tough	>3500	>800	Medium/low
Silicon nitride	hard/tough	>3500	>800	Medium/low
Metal matrix composites	-	>3500	>500	Medium/high
General purpose	-	variable	Variable	Medium/high

Mounting – The mounting operation accomplishes three important functions (1) it protects the sample edge and maintains the integrity of a materials surface features (2) fills voids in porous materials and (3) improves handling of irregular shaped samples, especially for automated sample preparation. The majority of metallographic sample mounting is done by encapsulating the sample into a compression mounting compound (thermosets - phenolics, epoxies, diallyl phthalates or thermoplastics - acrylics), casting into ambient castable mounting resins (acrylic resins, epoxy resins, and polyester resins), and gluing with a thermoplastic glues.

Compression mounting - For metals, compression mounting is widely used. Phenolics are popular because they are low cost, whereas the diallyl phthalates and epoxy resins find applications where edge retention and harder mounts are required. The acrylic compression mounting compounds are used because they have excellent clarity. Table IIIa and IIIb list the application as well as the physical properties of the most common compression mounting resins.

Table IIIa. Compression Mounting Application Properties

	Phenolics	Acrylics	Epoxy (glass filled)	Diallyl Phthalates
Cost	Low	Moderate	Moderate	Moderate
Ease of use	Excellent	Moderate	Good	Good
Availability of colors	Yes	No	No	No
Cycle times	Excellent	Moderate	Good	Good
Edge retention	Fair	Good	Excellent	Excellent
Clarity	None	Excellent	None	None
Hardness	Low	Good	High	High

Table IIIb. Compression Mounting Resin Physical Properties

Resin	Phenolic	Acrylic	Epoxy	Diallyl Phthalate
Form	Granular	Powder	Granular	Granular
Specific gravity (gm/cm^3)	1.4	0.95	1.75-2.05	1.7-1.9
Colors	Black, Red, Green	Clear	Black	Blue
Shrinkage (compression)(in/in)	0.006	-	0.001-0.003	0.001-0.003
Coefficient of Linear Thermal Expansion (in/in/°C 10^{-6})	50	-	28	19
Chemical resistance	Glycol, petrochemicals, solvents, some acids and bases	Alcohol, dilute acids & alkalies, and oxidizers	Solvents, acids, alkalies	Solvents, acids, alkalies
Molding temperature	150°-165°C (300°-330°F)	-	143°-177°C (290°-350°F)	160°-177°C (320°-350°F)
Molding pressure	21-28 MPa (3050-4000 psi)	-	17-28 MPA (2500-4000 psi)	24-41 MPA (3500-6000 psi)
Hardness	-	Rockwell M63	Barcol 72	-
Curing time (1/2" mount @ temp.) and pressure)	90-120 seconds	2-4 minutes	90-120 seconds	90-120 seconds

Castable Mounting Resins - Castable resins provide a convenient way to mount and orient specimens for microstructural preparation. Castable resins are monomer resins utilizing a catalyst or hardener for polymerization. Polymerization results in cross-linking of the polymer to form a relatively hard mount. Castable resins are advantageous because they do not require specialized mounting equipment. In addition, mounting high volumes of samples is faster then by compression mounting because multiple specimens can be mounted at the same time. Epoxy, acrylic and polyester resins are the most popular metallographic casting compounds.

Planar Grinding - or course grinding is required to planarize the specimen and to reduce the damage created by sectioning. The planar grinding step is accomplished by decreasing the abrasive grit/ particle size sequentially to obtain surface finishes that are ready for polishing. Care must be taken to avoid being too abrasive in this step, and actually creating greater specimen damage than produced during cutting (this is especially true for very brittle materials such as silicon).

The machine parameters, which effect the preparation of metallographic specimens, include grinding/polishing pressure, relative velocity distribution, and the direction of grinding/polishing.

Grinding Pressure - Grinding/polishing pressure is dependent upon the applied force (pounds or Newton's) and the area of the specimen and mounting material. Pressure is defined as the Force/Area (psi, N/m2or Pa). For specimens significantly harder than the mounting compound, pressure is better defined as the force divided by the specimen surface area. Thus, for larger hard specimens higher grinding/polishing pressures increase stock removal rates, however higher pressure also increases the amount of surface and subsurface damage. Note that for SiC grinding papers, as the abrasive grains dull and cut rates decrease, increasing grinding pressures can extend the life of the SiC paper. Higher grinding/polishing pressures can also generate additional frictional heat, which may actually be beneficial for the chemical mechanical polishing (CMP) of ceramics, minerals and composites. Likewise for extremely friable specimens such as nodular cast iron, higher pressures and lower relative velocity distributions can aid in retaining inclusions and secondary phases.

Grinding Direction - The orientation of the specimen can have a significant impact on the preparation results, especially for specimens with coatings. In general, when grinding and polishing materials with coatings the brittle component should be kept in compression. In other words, for brittle coatings the direction of the abrasive should be through the coating and into the substrate. Conversely, for brittle substrates with ductile coatings, the direction of the abrasive should be through the brittle substrate into the ductile coating.

Manual Preparation - In order to insure that the previous rough grinding damage is removed when grinding by hand, the specimen should be rotated 90 degrees and continually ground until all the scratches from the previous grinding direction are removed. If necessary the abrasive paper can be replace with a newer paper to increase cutting rates.

Automated Preparation - The key to successful automated preparation is to thoroughly clean the specimens between each abrasive grit size that is used. The tracking of the specimens should also uniformly break down the SiC paper, otherwise non-uniform grinding will occur (especially for hard specimens in soft mounts). In other words, the specimen should track across the entire diameter of the SiC paper.

GRIT NUMBER		
European (P-grade)	Standard grit	Median Diameter, (microns)
60	60	250
80	80	180
100	100	150
120	120	106
150	150	90
180	180	75
220	220	63
P240	240	58.5
P280		52.2
P320	280	46.2
P360	320	40.5
P400		35
P500	360	30.2
P600	400	25.75
P800		21.8
P1000	500	18.3
P1200	600	15.3
P2400	800	6.5
P4000	1200	2.5

Abrasive Selection - The most common abrasive used for planar grinding metal and polymer metallographic specimens is silicon carbide (SiC). Silicon carbide is an excellent abrasive for planar grinding because it is very hard and maintains a sharp cutting edge as it breaks down during cutting. SiC abrasives are typically listed by the grit size. Table VI shows the two most common grit size-numbering systems, along with their respective median diameter particle size.

Planar Grinding Recommendations –

Metallic Specimens - For metallic specimen grinding, sequential grinding with silicon carbide (SiC) abrasive paper is the most efficient and economical rough grinding process. Although extremely coarse grit abrasive papers can be found, it is recommended that a properly cut specimen not be rough ground with an abrasive greater than 120 grit SiC paper. A typical abrasive grinding procedure would consist of 120 or 240 grit SiC paper followed by decreasing the size of the SiC paper (320, 400, and 600 grit). Finer papers are also available for continued abrasive paper grinding (800 and 1200 grit).

In addition to the correct sequence and abrasive size selection, the grinding parameters such as grinding direction, load and speed can affect the specimen flatness and the depth of damage. The basic idea is to remove all of the previous specimen damage before continuing to the next step while maintaining planar specimens.

Electronic Specimens - Grinding electronic components is very dependent upon both the substrate (silicon, alumina, barium titanate, plastic PCB's, etc) and the metallic materials used. In general, when grinding plated or coated materials, it is recommended that the coating be prepared in compression to prevent the coating from separating from the substrate. Silicon specimens should be have been sectioned with a fine grit diamond blade and cut as near as possible to the area of interest. For rough grinding, fine abrasives such as 400 or 600 grit SiC or a 15-micron diamond lapping film is recommended to avoid producing more damage than created during sectioning. Hard ceramic substrates (especially porous materials) should be rough polished with diamond lapping films to minimize edge rounding and relief between the widely varying materials hardness'.

Plasma Spray Components - Similar to the preparation of electronic components, plasma spray coatings should be kept in compression to minimize the possibility of delamination at the coating/ substrate interface. For ceramic plasma spray coatings, diamond-lapping films are recommended to minimize edge rounding or relief and to maintain the integrity of any inclusions within the coating.
Ceramics and Ceramics Composites - Rough grinding ceramics and ceramic matrix composites should be performed with 15 or 30 micron diamond on a metal mesh cloth in order to obtain adequate stock removal and to minimize surface and subsurface damage.
Plastics and Plastics Composites - Plastics are generally very soft and therefore can be planar ground with sequentially decreasing SiC abrasive paper grit sizes. When plastics are used in conjunction with hard ceramics, planar grinding can be very tricky. For these composite materials cutting must minimize damage as much as possible because almost all grinding and polishing will cause relief between the soft plastic and the hard ceramic. Following proper cutting, grinding with as small as possible a diamond (6-micron diamond) on a metal steel mesh cloth or the use of lapping films is suggested.

Rough Polishing - The purpose of the rough polishing step is to remove the damage produced during cutting and planar grinding. Proper rough polishing will maintain specimen flatness and retain all inclusions or secondary phases. By eliminating the previous damage and maintaining the microstructural integrity of the specimen at this step, a minimal amount of time should be required to remove the cosmetic damage at the final polishing step. Rough polishing is accomplished primarily with diamond abrasives ranging from 9 micron down to 1-micron diamond. Polycrystalline diamond because of its multiple and small cutting edges, produces high cut rates with minimal surface damage, therefore it is the recommended diamond abrasive for metallographic rough polishing on low napped polishing cloths.

Final Polishing - The purpose of final polishing is to remove only surface damage. It should not be used to remove any damage remaining from cutting and planar grinding. If the damage from these steps is not complete, the rough polishing step should be repeated or continued.

Etching - The purpose of etching is to optically enhance microstructural features such as grain size and phase features. Etching selectively alters these microstructural features based on composition, stress, or crystal structure. The most common technique for etching is selective chemical etching and numerous formulations have been used over the years. Other techniques such as molten salt, electrolytic, thermal and plasma etching have also found specialized applications.
Chemical Etching - selectively attacks specific microstructural features. It generally consists of a mixture of acids or bases with oxidizing or reducing agents. For more technical information on selective chemical etching consult corrosion books which discuss the relationship between pH and Eh (oxidation/reduction potentials), often known as Eh-pH diagrams or Pourbaix diagrams.

Molten Salt etching is a combination of thermal and chemical etching techniques. Molten salt etching is useful for grain size analysis for hard to etch materials such as ceramics. The technique takes advantage of the higher internal energy associated at a materials grain boundaries. At the elevated temperature of molten salts, the higher energy at the grain boundaries is relieved, producing a rounded grain boundary edge, which can be observed by optical, or electron microscope techniques.

Thermal etching is a useful etching technique for ceramic materials. Thermal etching is technique that relieves the higher energy associated at the grain boundaries of a material. By heating the ceramic material to temperatures below the sintering temperature of the material and holding for a period of minutes to hours the grain boundaries will seek a level of lower energy. The result is that the grain boundary edge will become rounded so that it can be evaluated by optical or electron microscope techniques.

Depending upon the ceramic material, the atmospheric condition of the furnace may need to be controlled. For example, etching silicon nitride will require either a vacuum or an inert atmosphere of nitrogen or argon to prevent oxidation of the surface to silicon dioxide.

Plasma etching is a lesser-known technique that has been used to enhance the phase structure of high strength ceramics such as silicon nitride. For silicon nitride, the plasma is a high temperature fluoride gas, which reacts with the silicon nitride surface producing a silicon fluoride gas. This etching technique reveals the intragrain microstructure of silicon nitride.

Metallographic Analysis –

Metallurgical analysis (metallography) of the microstructural provides the Material Scientist or Metallurgist information varying from phase structure, grain size, solidification structure, casting voids, etc. Figure 2 shows an example of two cast iron structures. Figure 2a shows a cast iron microstructure which has graphite flakes. Over time this materials will most likely fail under load. On the other hand by adding some solidifying agents the cast iron can be made to form the more durable graphite nodules (Figure 2b).

Figure 3 shows the grain size of a tough pitch copper. Analysis of a materials grain size provides valuable information regarding a materials physical hardness and ductility. Microstructural analysis can also provide very useful information about the types of phases that occur during cooling. Figure 4 shows the dendritic growth of the microstructure for an aluminum-silicon alloy which formed during solidification. The direction and size of these dendrites again relate to the materials strength and durability.

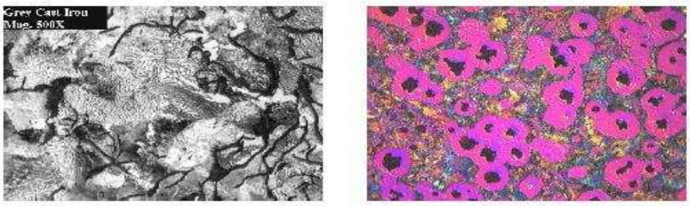


Figure 2. Cast Iron microstructures (a) graphite flakes (b) nodules

Figure 3. Tough Pitch Copper microstructure showing grain size

Figure 4. Aluminum-silicon alloy solidification microstructure

Other examples of metallographic analysis include: analyzing printed circuit board (PCB) solder hole connections, microelectronic failure analysis, steel heat treating such as surface hardening, ceramic grain size and porosity, polymer composite fiber wetting and orientation, etc.

Microscopic Analysis

Bright Field (B.F.) illumination is the most common illumination technique for metallographic analysis. The light path for B.F. illumination is from the source, through the objective, reflected off the surface and returning through the objective and back to the eyepiece or camera. This type of illumination produces a bright background for flat surfaces with the non-flat features (pores, edges, etched grain boundaries) being darker as light is reflected back at an angle.

Field (D.F.) illumination is a lesser know illumination but powerful illumination technique. The light path for D.F. illumination is from the source, down the outside of the objective, reflected off the surface and returning through the objective and back to the eyepiece or camera. This type of illumination produces a dark background for flat surfaces with the non-flat features (pores, edges, etched grain boundaries) being brighter as light is reflected at an angle back into the objective.

Differential Interference Contract (DIC) - is a very useful illumination technique for providing enhanced specimen features. DIC uses a normarski prism along with a polarizer in the 90° crossed positions. The two light beams are made to coincide at the focal plane of the objective, thus rendering height differences visible as variations in color.

Optical Interferometry - is a technique that provides precise details of a materials surface topography. The simplest interferometer employs the interference between two beams of light. One beam is focused on the specimen and the second beam on an optically flat reference surface. The two reflected beams are then recombined by the beam splitter and pass through the eyepiece together. The two beams reinforce each other for those points on the specimen for which their path lengths are either the same of differ by an integral multiple of the wavelength, $n\lambda$. The beams cancel for path differences of $n\lambda /2$. Today's interferometers provide quantitative 3-dimensional surface topography information.

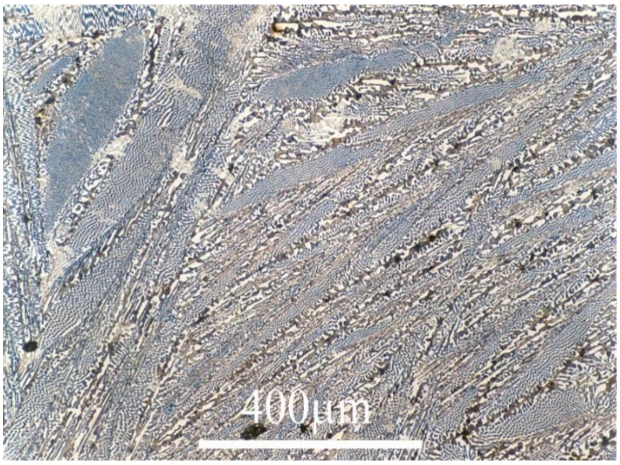
The two most common optical interferometry techniques include - phase shifting interferometry (PSI) which uses a PZT to shift the optical path of the objective and vertical scanning interferometry (VSI) which changes the focus range of the objective. The main differences between PSI and VSI are that PSI has a higher z-axis resolution, whereas VSI has a larger scan range. PSI and VSI can also be combined to provide both high surface resolution and a larger scan range.

Atomic Force Microscopy (AFM) - Is a non-optical technique that provides angstrom level surface profiling information. AFM's are very powerful tools and can be provide electrochemical, magnetic, and other surface feature information.

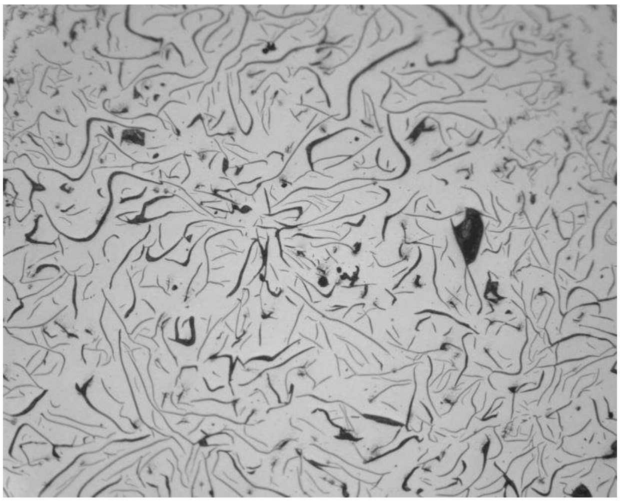
References:

1. http://www.lt-scientific.com/me_c_cmr.pdf
2. www.metallographic.com
3. http://www.lapmaster.com/metallographic-products/abrasive-cut-off-wheels/default.html
4. http://dce.karabuk.edu.tr/metalografik.html
5. http://www.tse.org.tr/hizmetlerimiz/deney-hizmetleri/laboratuvarlar/makine-laboratuvar%C4%B1-(gebze)/sertlik-ve-metalografik-analizler-b%C3%B6l%C3%BCm%C3%BC
6. http://imechanica.org/files/handout9.pdf

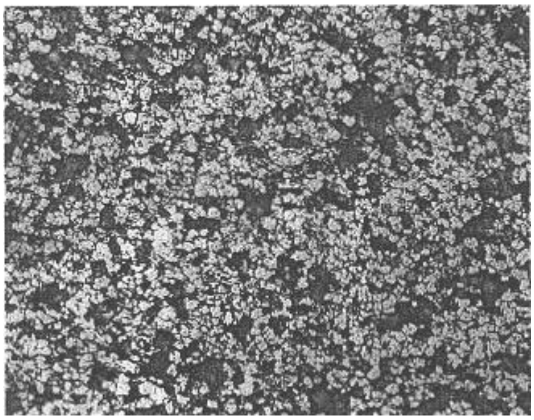
Pictures from different structures



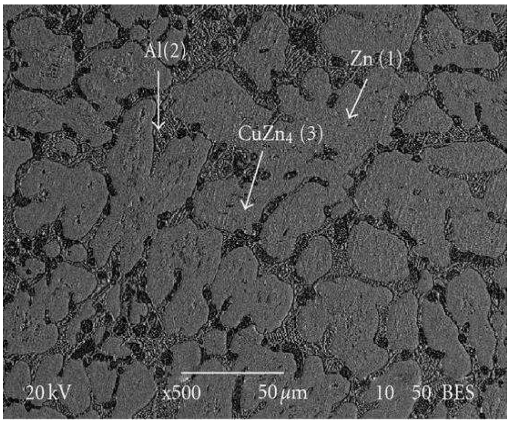
Ledeburite is a mixture of 4.3% carbon in iron and is a eutectic mixture of austenite and cementite. Because of mixture, there occur some gray waves.



Gray cast iron is a kind of cast iron that has graphitic structure. Chemical composition of structure should be 2.5-4.0% carbon and 1-3% silicon.



Al-Si alloy. High levels of silicon (4.0–13%) contribute to give good casting characteristics. Aluminium alloys are widely used in engineering structures and components where light weight or corrosion resistance is required.

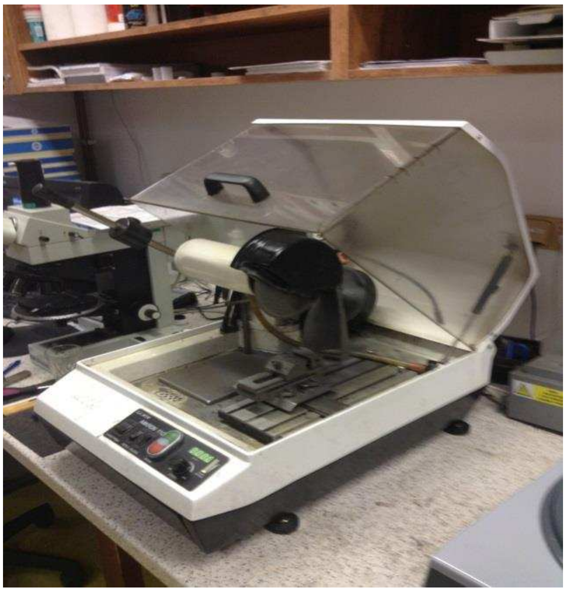


Cu-Zn alloy. Copper alloys are metal alloys that have copper as their principal component. They have high resistance against corrosion. The most known copper alloy is bronze and brass.

Practical Part

<u>Day 1 (June 12th, 2014)</u>

All the students chose their own samples to examine by pursuing metallographic traditions step by step under supervising of Ing.Svobodova and her laboratory colleagues. Firstly, I picked $AlMg_3$ for my sample. Then, using the abrasive disc in the cutting machine, cut the sample twice for having proper material shape to examine at it.



After obtaining proper shapes, it is time to put the sample in resin. But first we need to prepare it. First, the sample has to be cleaned by water to put it in the plastic mold.

Then, in a different cup, we prepare resin casting. The things that we use is a chemical called Dentacryl Technicky. When we research, we find out that it is a methylmethacrylate compound. We add liquid and powder version of this compound altogether and try to obtain proper concentration.

After that pour it to plastic molds that has our sample. We need to care that the surface we wish to examine should be upside down to the bottom so that after taking out of sample from mold we can carry out some procedure on it.

After pouring it down, we have to wait 24 hours so that solidification process can be completed.

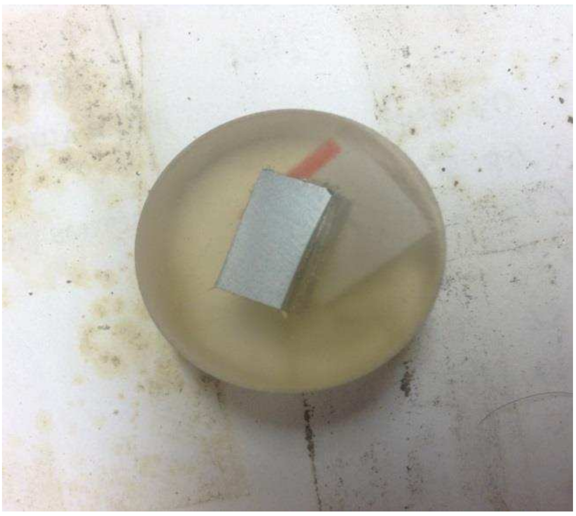
Day 2 (June 24th, 2014)

We take the sample off from the plastic mold so that we can do operations on it. So, we turn the centrifugal machine on and put the proper disc on it. First, we place the abrasive paper with number p280. When it starts turning, we put our sample on turning disc with the abrasive paper. Here the grinding starts. When we need more power, we change the abrasive paper. Respectively; p280, p800, p1500, p2500. The number of paper goes up, the size and number of grit on the paper gets increased.

Day 3 (June 25th, 2014)

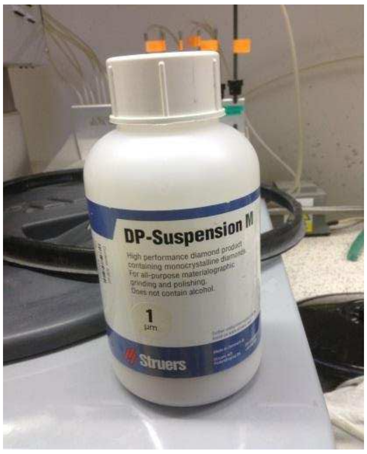
Today we start to polish our sample. For that, we change disc and put the one provides our needs. We prefer a disc that has some roughness over the surface. We add AlO_2 for lubricating. The reason why we polish is to take the damage and scratches away that occured during cutting the material. The surface will be flat and without scratches. In this way, we will have more clear and specific results over the surface. But my sample looks quite scratchy in comparison with other samples. It requires so much time to fix its surface more than we do. There are a lot of bulges and scratches. Despite of polishing hardly and repeatedly it is not bright and shiny.

After polishing, we wash and add alcohol and dry. Then, send it to microscope. But my sample $AlMg_2$ does not look smooth and sharp.



<u>Day 4 (June 26th, 2014)</u>

Today we will work on final polishing. We will use 3 different lubricants so that surface can be shiny enough and scratch free. Firstly, we use green suspension called 3 µm -monocrystalline diamond suspension. We put another disc. Today we will use 3 different discs. All of them different roughness values. And they started to have softy surfaces. After spraying the 3 µm suspension, we press the sample over the disc. But my sample does not respond as usual. We check it with microscope.

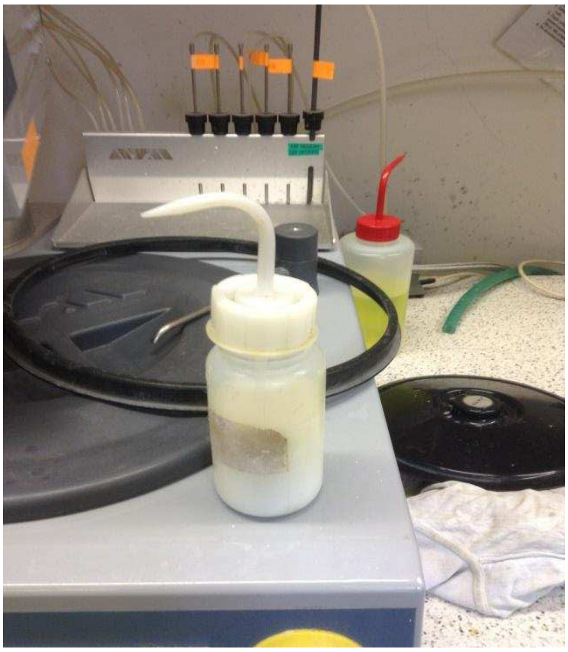


Then, we change the disc and add another suspension called 1 µm – DP-Suspension M. We press the sample over the disc but my sample still does not respond. Its surface still same but started to be shiny. Not so much as other samples but when we compare its situation in respect with first polishing.

After washing, alcoholizing and drying, checked it with microscope. It seemed worse in respect with first polishing.

Keeping polished by the disc does not work with my sample at all.

For the last, we used some other nameless suspensions or lubricants yellow and white colored. Of course, changed the disc for the last time. Add lubricant and put pressure over the sample. After washing, alcoholizing and drying, checked it with microscope. It still seems bad.



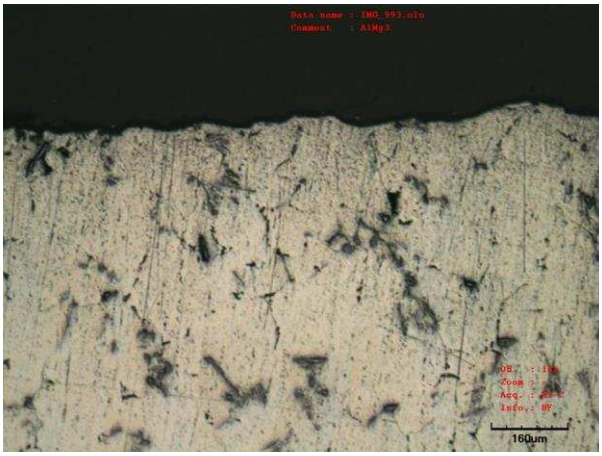
Day 5 (June 27th, 2014)

Today, Ing.Svobodova is going to take microscobic pictures using € 125.000 – microscobe called **Olympus Lext, OLS 3100**.

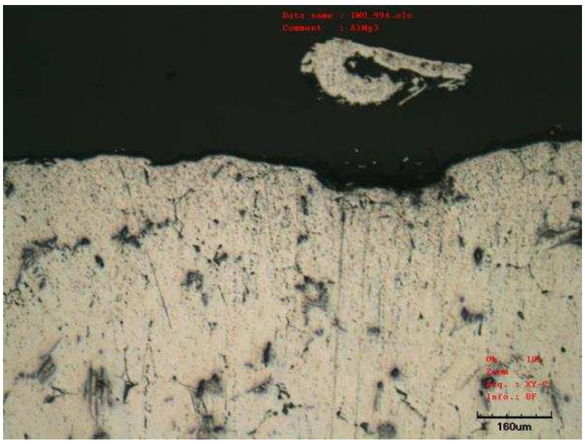
It provides pictures in high resolution or 3D. It depends what you will need. Its features are:

1. Ideal 3D imaging achievable with a single click.
2. Industrial-leading resolution and repeatability for extremely reliable measurement data.
3. Ability to observe and analyze all types of specimens thanks to versatile observation methods and a variety of 3D image presentation patterns.

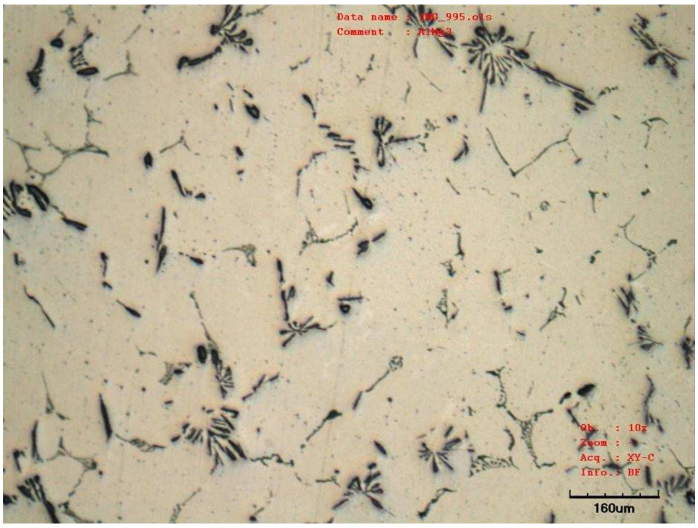
Here are some images of metallographic analysis.



When we look the image, we can see some pitting corrosion. But in comparison of other samples this corrosion is not much and harmful. Material is highly corrosion resistant. It is a useful property for manufacture. Scratches occured because of grinding direction.



This part of sample is very corroded. There scratches over the surface and dendrites. Because of them we cannot have a good examination over the surface. But we can learn the negative sides of metallograhic images for how cannot we study over a surface? It can answer this question surroundly.



We can see a lot of dendrites all over the surface. It is a problem for sample. Because dendrites are uniform and solid, no way to split them up. So it makes a big problem to solve for whole body of sample. It occurs because of undercooling or supercooling of molten material.

And about the black dots, they are alpha or betha phases which means Al solves Mg or backwards. So that the dots occur.

Properties of $AlMg_3$


ALUMINIUM CENTRUM s.r.o.
Your expert in aluminium

Mechanical qualities according to EN 573 - 3 (DIN 1725)

1050A (Al99,5) 5754 (AlMg3) 6060 (AlMgSi0,5) 6082 (AlMgSi1) 2017A (AlCuMg1) 7075 (AlZnMgCu1,5
5005 (AlMg1) 5083 (AlMg4,5Mn) 6061 (AlMg1SiCu) 2007 (AlCuMgPb) 2024 (AlCuMg2)

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AlMg3

Euro standards	AW-5754
EN marking	AW-AlMg3
DIN material number	3.3535
DIN marking	AlMg3
Condition	0
Tensile strength RM (Mpa)	min. 190 max.240
Yield limit / point / R p0,2	-
Tensibility	min. 80
Hardness (HBS)	52
Electric conductivity	20 - 23
Thermal conductivity	140 - 160
Density / Specific weight	2,66
Surface treatment ability	good
Surface finishing	satisfactory
Weld ability	good
Corrosion resistance / Sea water resistance	very good - good
Workability	satisfactory